Build Proof — relay_v2.kicad_pcb (placement engine v2)

Board: 4-channel relay training board, 100×60mm, 36 footprints, built 2026-06-11 via `build_pcb(placement_mode="v2")`. Every row below was computed by deterministic code (`placement_v2/ledger.py`); "done" is defined as `all_green() == True`, never by prose. Gate A re-derives every constraint from the written .kicad_pcb on disk. Gate B verdicts come from an adversarial vision review over the 4-view render standard, parsed through a strict JSON schema (prose answers are rejected).

Stage ledger — ALL GREEN: True

Stage	Verdict	Checks run	Violations	in/out SHA-256	Detail
certify	**PASS**	certificates x32	—	`97cd162cbbb1f066…` `d9b6472602a62aca…`	
cells	**PASS**	relay_driver:K1: pin_attach x5, sequences x0, zero_courtyard_overlap relay_driver:K2: pin_attach x5, sequences x0, zero_courtyard_overlap relay_driver:K3: pin_attach x5, sequences x0, zero_courtyard_overlap relay_driver:K4: pin_attach x5, sequences x0, zero_courtyard_overlap single:C1: singleton single:C2: singleton single:J1: singleton single:J2: singleton single:J3: singleton single:J4: singleton single:L1: singleton single:L2: singleton	—	`97cd162cbbb1f066…` `3ab3591ad6351667…`	12 cells (4 subcircuits detected)
floorplan	**PASS**	groups x1 cells x12	—	`97cd162cbbb1f066…` `0c40d11988c141c5…`	board 100.0x60.0mm
gate_a	**PASS**	pin_attach x20 sequence x2 edge_pin_distance x4 edge_pin_face_out x4 (skipped: orientation check not implemented) contain x36 courtyard_pair x630 keepout x0 isolation_pair x0	—	`cdc1af135ae1383a…` `cdc1af135ae1383a…`	re-derived from .kicad_pcb on disk
gate_b	**PASS**	bodies_no_overlap bodies_flat_on_board bodies_on_pads orientation_sane antenna_zone_clear groups_visually_coherent	—	`cdc1af135ae1383a…` `2d:1f4715559f2a,…`	6 verdicts over 4 views

What each stage proves

Stage	Guarantee
certify	Every footprint matches its hash-keyed certificate; a drifted footprint (changed pads) halts the build before placement.
cells	Each subcircuit laid out by pin connectivity proved its internal constraints (pin-attach distances, sequences, zero courtyard overlaps) before being frozen as a rigid cell.
floorplan	Cells packed with hard constraints: board containment, no overlaps, terminals pitch-locked under relays, connector strip outermost and edge-snapped; legalization raised no residuals.
gate_a	The WRITTEN FILE re-parsed from disk satisfies the full constraint IR: 20 pin attachments, 2 sequences, 4 edge pins, 36 containments, 630 courtyard pairs — zero violations.
gate_b	Adversarial vision over 4 views: no body overlaps, all bodies flat and on their pads (after the −1.8mm relay model calibration), terminals opening outward, channels periodic. 6/6.

Render hashes bound to this exact board: 2d:1f4715559f2a,3d_top:7a013f2aebe3,3d_iso:6ae815f27b0b,3d_iso_back:2d1d60c4ae0f